

Generation Interconnection System Impact Study Report

For Queue Project Q21-1937

Mill C 345 kV
October 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1937. The Interconnection Customer proposes a 76.0 MW battery energy storage facility interconnecting at the Mill C 345 kV substation in the New Jersey area. The request was received on November 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 501 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$865,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FELIDA to WEST GALLATIN 345 kV	WALLENPAUPACK 500/345 kV TX	107.0	1,858	10.7
CLEAR SPRINGS 500/345 kV XFMR	BURNS to HILLS CHAPEL 345 kV line	105.4	1,702	11.9
DRISCOLL to JACKASS FLATS 138 kV	BURNS to HILLS CHAPEL 345 kV	102.6	269	5.1
CLEAR SPRINGS to WALNUT 345 kV	WALLENPAUPACK 500/345 kV Auto	101.2	1,676	10.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST GALLATIN to FELIDA 345 kV	Reconductor line	\$14,613,000	1% (\$167,000)
CLEAR SPRINGS 500-345 kV TX	Replace transformer with higher-rated unit	\$27,552,000	1% (\$313,000)
DRISCOLL to JACKASS FLATS 138 kV	Reconductor line	\$567,000	36% (\$202,000)
WALNUT to CLEAR SPRINGS SUBSTATION 345 kV	Reconductor line	\$16,075,000	1% (\$183,000)
	Total allocated to project		\$865,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.